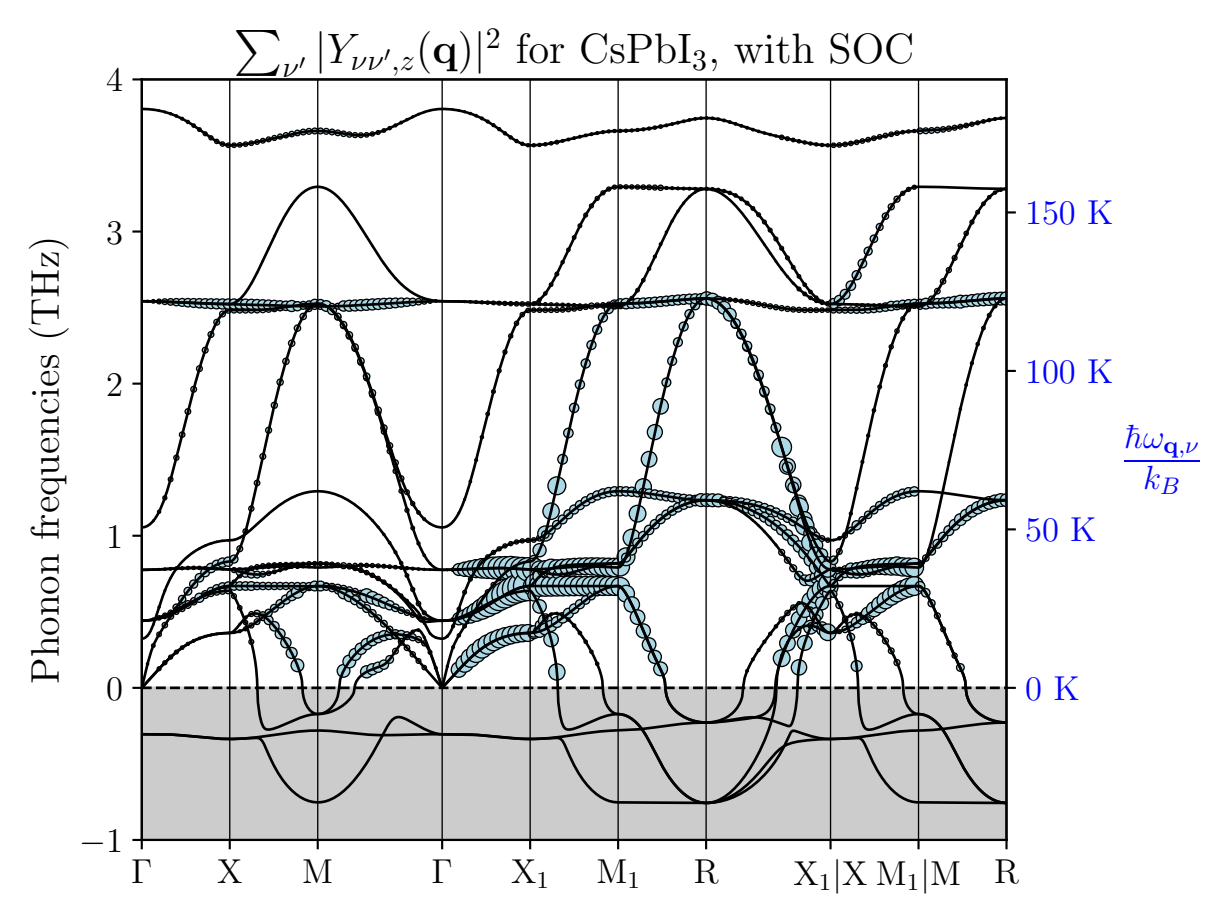

$\sum_{\nu'} |Y_{\nu\nu',z}(\mathbf{q})|^2$ for $CsPbI_3$, with SOC
Phonon frequencies (THz)
4
3
2
1
0
−1
Γ
X
M
Γ
X_1
M_1
R
$X_1|X$
$M_1|M$
R
150 K
100 K
50 K
0 K
$\frac{\hbar\omega_{\mathbf{q},\nu}}{k_B}$